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MORE MOORE

1. INTRODUCTION

System scaling enabled by Moore's scaling is increasingly challenged by the scarcity of resources such as power and data bandwidth. This has become more challenging under the requirements of seamless interaction between big data and instant data (Figure MM-1). Instant data generation requires ultra-low-power devices with an "always-on" feature at the same time as high-performance devices that can generate the data instantly. Big data requires abundant computing, communication bandwidth, and memory resources to generate the service and information that clients need. Need for data next to compute accelerated as AI workloads increased.

The More Moore International Focus Team (IFT) of the International Roadmap of Devices and Systems (IRDS) provides physical, electrical, and reliability requirements for logic and memory technologies to sustain power, performance, area, cost (PPAC) scaling for edge and cloud applications. This is done over a time horizon of 15 years for mainstream/high-volume manufacturing (HVM).



Figure MM-1 Big data and instant data.

Following systems drivers are forecasted to impact the More Moore logic technologies:

Mobile

- Heterogenous integration
- Edge computing
- Extreme reality (VR/AR)
- AI enhanced edge computing and connectivity (mobile phone, 6G, cellular, IoT, Wi-Fi, wireless connectivity, smart cameras, and speakers) driving any data, any place, highest speed, and lowest power with content-rich data.

Data and HPC servers – cache integration, memory, IO

- AI accelerators in enterprise/cloud
- Codec ASICs - 24/7/365 continuous run of video and audio (codec), 5 years minimum time
- Networking – Always-on, 500W power envelope
- Advanced driver assistance system (ADAS) chips – Autonomous driving
- Memory and IO solutions for AI, graphics, HPC

Novel compute fabrics

- Neural processing unit

- Fine-pitch 3D stacking
- Reconfigurable compute fabrics
- Smart 2.5D interposers

1.1. CURRENT STATE OF TECHNOLOGY

A major portion of semiconductor device production is devoted to digital logic that needs to support a technology platform for two device types: 1) high-performance logic, and 2) low-power/high-density logic. Key considerations for this technology platform are speed, power, density, cost, capacity, and time-to-market. The More Moore roadmap provides an enablement view for continued scaling of MOSFETs to maintain historical trends of improved device performance at reduced power and cost and at high volume [1]. 2025 is the 100th anniversary of the concept of the Field Effect Transistor (FET), which was first captured as a patent filed by Julius Lilienfeld on October 22, 1925. While reaching the 100th anniversary of transistor, we have seen many critical enablers for the continuous enhancement of transistor on top of geometric scaling, such as strained silicon, high-k metal gate, raised source drain, multi-gate transistor such as finFET, and now evolving to gate-all-around devices.

1.2. DRIVERS AND TECHNOLOGY TARGETS

The following applications drive the requirements of More Moore technologies that are addressed in the IRDS [2]:

- High-performance computing—more performance at constant power density (constrained by thermal)
- Mobile computing—more performance and functionality at constant energy (constrained by battery) and cost
- Autonomous sensing and computing (IoT)—targeting reduced leakage and variability

Technology drivers include the following focal items:

- Logic technologies
- Ground rule scaling
- Performance boosters
- Performance-power-area (PPA) scaling
- 3D integration
- Memory technologies
- DRAM technologies
- Flash technologies
- Emerging non-volatile-memory (NVM) technologies

More Moore targets bringing PPAC value for node scaling every 2–3 years [3]:

- (P)erformance: >10% more operating frequency at scaled supply voltage
- (P)ower: >20% less energy per switching at a given performance
- (A)rea: >30% less chip area footprint
- (C)ost: <30% more wafer cost – 15% less die cost for scaled die.

System scaling considers the co-integration of logic, memory and IO solutions bringing the following targets:

- TOPS (Tera Operations per Second): Throughput
- TOPS/W (TOPS per Watt): Energy efficiency
- TOPSxTOPS/W/Area is an indicator of Energy-Area-efficient performance (aka EDP: Energy Delay Product at Unit Area)
- 2.0-2.4x scaling of TOPSxTOPS/W/Area for node scaling per frame, per inference, per training, and/or per pocket

These scaling targets have driven the industry toward a number of major technological innovations, including material and process changes such as high- κ gate dielectrics and strain enhancement, and in the near future, new structures such as gate-all-around (GAA); alternate high-mobility channel materials, wafer backside processing (e.g. backside PDN) [4], and new

4.2.2. Starting Substrate and high-mobility channels

Bulk silicon will still remain the mainstream substrate while silicon-on-insulator (SOI) and strain-relaxation-buffer (SRB) will be used to support better isolation (e.g., RF co-integration) and defect-free integration of high-mobility channels, respectively.

High-mobility materials such as Ge and III-V bring promise in increasing drive current by means of an order of magnitude increase in intrinsic mobility. It is likely that high-mobility channels will occur in the form of chiplets dedicated to high-performance functions, such as high-speed IOs, high-current analog drivers, RF, photonics devices, power management, etc, which does not need to follow aggressive dimensional scaling. Improved performance and enablement of new features in the total system needs to be weighed against the cost, determined by substantial investment in new tools and fab infrastructure. On the other hand, increasing the number of vertical stacked nanosheets employing high mobility channels allow very high performance at reduced footprint [18].

4.2.3. Strain engineering

Strain engineering has been used as one of the most effective solutions in the last decade, as illustrated for the 32nm node and earlier. However, the effect of those stressors may not extrapolate intuitively into newer nodes as a result of new device architectures and 3D device schemes. With the scaling down of gate pitch, SiGe on the source/drain epitaxial (S/D EPI) contact and strain relaxation buffer (SRB) remain as effective boosters to scale mobility more than double on top of high-mobility channel material. SiGe channel for PMOS and strained Si channel for NMOS has been successfully demonstrated on a 7nm CMOS platform using SRB and on gate-all-around devices. Other strain engineering techniques also contain gate stressor and ground plane stressors, which adopt the beneficiary vertical stress components for NMOS.

4.2.4. Reducing parasitic device resistance

Controlling source/drain series resistance within tolerable limits will become increasingly difficult. Due to the increase of current density, the demand for lower resistance with smaller dimensions at the same time poses a great challenge. It is estimated that in current technologies, series resistance degrades the saturation current by 40% or more. External resistance impact on the drive current is expected to become worse with the gate pitch scaling. In addition, increasing interconnect resistance by scaling is expected to necessitate much lower resistance values for the device contact and therefore more emphasis is given to increase the contact surface area by using the wrap-around contact schemes.

4.2.5. Reducing parasitic device capacitance

Parasitic capacitance between gate and source/drain terminal of the device is expected to increase with technology scaling. In fact, this component is getting more important than channel-capacitance-related loading whenever the standard cell context is considered and even more elevated in the GAA structures as a result of unused space between stacked devices. There is a need to focus on low- κ spacer materials and integration of inner spacer schemes in the GAA device is necessary to block the gate di-electric interface facing the source/drain at the unused space of GAA device. Those schemes still need to provide good reliability and etch selectivity for S/D contact formation [19]. Also, there are significant limits in increasing finFET or lateral GAA device AC performance by increasing the height of the device (fin/nanosheet stack). Energy per switch vs. delay relationship seems to quickly saturate and then decline with increasing device height, requiring to reduce the vertical space between the GAA sheets. Scaling trend of key parasitic improvements is shown Figure MM-4.

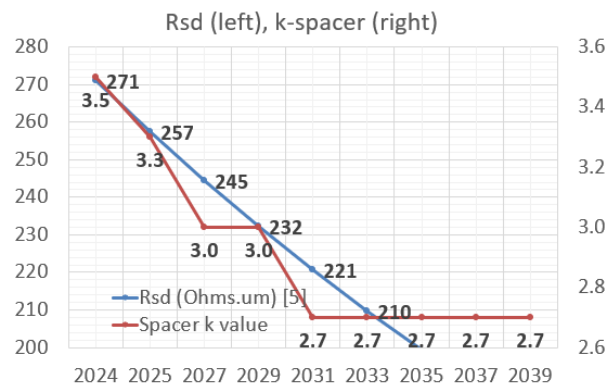


Figure MM-6 Scaling trend of device S/D access resistance (R_{sd}) and k -value of device spacer. Note [5]: R_{sd} is the total parasitic series resistance (source plus drain) per micron of MOSFET width. These values include all

components such as accumulation layer, spreading resistance, sheet resistance, and contacts. It is assumed that there is 5% improvement per node cycle (2 years or 3 years).

4.2.6. Increasing drive per footprint

Lateral GAA devices enable a higher drive at unit footprint (by enabling drive in the third dimension) if device pitch can be aggressively scaled while increasing the number of stacked GAA devices [19]. This will then increase drive at unit footprint but bringing a trade-off between fringing capacitance between gate and contact, and series resistance. For the GAA devices this is accomplished by the increasing the number of GAA sheets while reducing the width of each GAA sheet. Complementary FET (CFET) will also scale the drive per footprint by increasing the number of devices as in GAA. This will substantially increase the amount of devices at unit footprint.

4.2.7. Improving electrostatics

LGAA devices bring better electrostatics than finFET by providing a gate control from all-sides of device channel. Since the devices are stacked on top of each other, the spacing between the devices needs to be kept smaller to reduce the parasitic capacitance between the source/drain and gate while still having adequate space of the gate dielectric and V_t -tuning work function metals deposition. Junction formation engineering, EOT scaling and density of interface traps (D_{it}) reduction are also needed to maintain the electrostatics control in the channel.

4.2.8. Improving device isolation

Besides the channel leakage induced by electrostatics, there are potentially other leakage sources such as sub-fin leakage or punchthrough current. This leakage current flows through the bottom part of the fin and the bottom GAA device from source to drain. Ground plane doping and bottom dielectric isolation below the channel would potentially solve this leakage problem.

4.2.9. Reducing process and material variations

Reducing variability would further allow supply voltage (V_{dd}) scaling. Controlling channel length and channel thickness are important to maintain the electrostatics in the channel. This would require, for example, controlling the profile of the fin and lithography processes to reduce the CD uniformity (CDU), line width roughness (LWR), and line edge roughness (LER). Dopant-free channel and low-variability work-function metals would reduce the variations in the threshold voltage. Gate stack passivation might also be needed to reduce the interface-related variations as well as maintaining the electrostatics and mobility.

4.2.10. Beyond CMOS for application-specific/complimentary functions

MOSFET scaling will likely become ineffective and/or very costly for the complementary SoC functions, such as memory selector, power header/footer switch, cross-bar switch, etc. Completely new, non-CMOS types of logic devices and maybe even new circuit architectures are potential solutions (see the Beyond CMOS chapter for detailed discussions). Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices, such as memories, onto the same chip. Even early adoption of Beyond CMOS technology and/or computing are likely to be adopted around 2029 by ferroelectric-FET, BEOL oxide transistors, IGZO, and/or 2D materials for ultra-low power applications and also memristors for neuromorphic applications [20].

4.3. PERFORMANCE-POWER-AREA (PPA) SCALING

The projected roadmap for the electrical specifications of logic core device is listed in Table MM-9. This edition of More Moore roadmap includes both logic and analog specifications of More Moore platform device. Analog specifications are derived from the device targets of a logic device but this would potentially require relaxation of contacted poly pitch on the same wafer to allow longer channel lengths. There would also be considerations such as reliability and matching where performance targets need to be derated in an effort to meet those concurrent goals, e.g. increasing overdrive voltage by stacking devices.

Table MM-9

Projected Electrical Specifications of Logic Core Device

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic Industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	"3nm" Enhanced Stacking	"2nm" Stacking	"1.4nm" Stacking	"A10 eq" Stacking	"A7 eq" Stacking	"A5 eq" Stacking	"A3.5 eq" 3DVL SI	"A2.5 eq" 3DVL SI	"A1.8 eq" 3DVL SI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
LOGIC DEVICE ELECTRICAL SPECS									
Power Supply Voltage - V _{dd} (V)	0.70	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
Subthreshold slope (mV/dec) - HP (mV/dec)	82	72	70	70	70	70	70	70	70
Subthreshold slope (mV/dec) - HD (mV/dec)	75	67	67	65	65	65	65	65	65
Capacitive equivalent thickness (CET) (nm) [2]	1.00	1.00	0.90	0.90	0.90	0.90	0.90	0.90	0.90
V _{t,sat} at I _{off} =10nA/um - HP (mV)	156	165	164	152	164	164	158	151	146
V _{t,sat} (mV) at I _{off} =100pA/um - HD (mV) [3][4]	288	271	276	260	266	265	261	257	257
Average effective mobility of N/P-MOS (cm ² /V.s)	125	100	80	80	60	60	60	60	60
R _{sd} (Ohms.um) [5]	271	257	245	232	221	210	199	189	180
Ballisticity Injection velocity (cm/s)	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06
V _{dsat} (V) - HP	0.092	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
V _{dsat} (V) - HD	0.104	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
Average I _{on} (uA/Perimeter Weff-um) of N/P-MOS at I _{off} =10nA/um - HP [6]	874	787	759	797	768	800	827	858	885
Average I _{on} (uA/sheet or fin) of N/P-MOS at I _{off} =10nA/um - HP [7]	88	170	158	115	161	139	119	98	85
Average I _{on} (uA/Perimeter Weff-um) of N/P MOS at I _{off} =100pA/um - HD [8]	644	602	546	589	565	595	614	635	646
Average I _{on} (uA/sheet or fin) of N/P MOS at I _{off} =100pA/um - HD [9]	65	130	114	85	119	104	88	72	62
C _{ch} total (fF/um ²) - HP/HD [8]	34.52	34.52	38.35	38.35	38.35	38.35	38.35	38.35	38.35
Gate height over fin (nm)	20	15	10	10	10	10	10	10	10
C _{ch} (fF/um) - HP [8]	0.44	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
C _{ch} (fF/um) - HD [8]	0.50	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
CV/I (ps) - FO3 load, HP [9]	1.06	0.96	0.87	0.83	0.86	0.69	0.67	0.64	0.62
I/CV (1/ps) - FO3 load, HP [10]	0.94	1.04	1.15	1.20	1.16	1.45	1.50	1.55	1.60
Energy per switching [CV2] (fJ/switch) - FO3 load, HP	0.65	0.49	0.40	0.40	0.40	0.33	0.33	0.33	0.33

An important speed metric for the transistor is the intrinsic delay (CV/I) where C includes the gate capacitance plus the gate fringing capacitances. These fringing capacitances have been found to be larger than the intrinsic capacitance over the channel region. This requires a modeling of parasitic components in the device. The ratio of total fringing capacitances to the gate capacitance over the channel is increasing with scaling.

In order to capture the behavior of a wireloaded datapath to connect the device parameters to SoC, we use a ring-oscillator-based circuit model where each stage is implemented with a D4 inverter driving a wireload with its branches driving three D4 inverters for the high-performance devices.

In this datapath model the delay of each stage is approximated by the Elmore expression given below [3]:

$$T_{del} = 0.69 \cdot R_{dr} \cdot C_{int} + (0.69 \cdot R_{dr} + 0.38 \cdot R_w) \cdot C_w + 0.69 \cdot (R_{dr} + R_w) \cdot C_{out}$$

where R_{dr} is the resistance of driver, C_{int} is the capacitance seen at the output of driver, R_w is the wire resistance, C_w is the wire capacitance, and C_{out} is the load capacitance due to the gates connected to the load. For logic technologies beyond 10nm the dominant term is typically found to be $R_w \cdot C_{out}$ [3]. This means that increasing the driver strength does not help if there is no improvement in the parasitic resistance of interconnect and/or a reduction in the parasitic loading of standard cell.

Projected standard cell and bitcell layout characteristics (e.g., number of active devices, W_{eff} , etc) are shown in Table MM-10.

Performance scaling across six nodes spanning from 2024 to 2039 is projected to have a mild increase (10-15% on a node-to-node basis) for datapaths with wireload resistance slowing down the performance scaling. We also take into account the wirelength reduction as function of area scaling translating into the reduction of wire-related loading capacitance and resistance. Wirelength is expected to further reduce as a result of 3DVL SI after 2031.

Energy per switching reduction is forecasted to become limited. This is mostly achieved by fin/device depopulation, which also enables the cell height reduction bringing a scaling of wire and cell related capacitances. We also consider that DT CO constructs such as contact-over-active, single diffusion break, tighter spacing between N and P, etc., as described in [6], will further reduce the standard cell height and width. After 2031 3DVL SI monolithic/sequential integration (full-scale 3DVL SI) would further maintain the scaling of the number of functions per unit cube.

Table MM-10 Logic cell and bitcell architecture.

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic Industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	Stacking	Stacking	Stacking	Stacking	CFET	CFET	CFET 3D	3DVL SI	3DVL SI
Backside structure options	finFET	Backside via	Direct contact	Decap-ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC CELL AND BITCELL ARCHITECTURE									
SRAM bitcell area (μm^2)	0.018	0.015	0.014	0.013	0.008	0.008	0.007	0.007	0.007
SRAM bitcell area portion	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
HD SRAM area density (Mbits/mm ²)	33	40	43	45	77	77	81	81	81
NAND2 active devices per PUI/PD for HP	2	1	1	1	1	1	1	1	1
NAND2 active devices per PUI/PD for HD	1	1	1	1	1	1	1	1	1
Power delivery structure	Frontside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via	Backside via
Power rail width (nm)	24	22	22	20	18	16	14	14	14
Cell height (nm) - HP	192	144	144	120	108	96	84	84	84
Cell height (nm) - HD	160	120	96	80	63	56	49	49	49
Cell height in 1 μm routing tracks - HD	5.00	5.00	4.00	4.00	3.50	3.50	3.50	3.50	3.50
Routing utilization target	0.80	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
NAND2-eq gate density (Mgates/mm ²) - HP	16	23	23	29	33	37	45	45	45
NAND2-eq gate density (Mgates/mm ²) - HD	19	27	34	43	57	64	76	76	76
Effective drive width of D4 ND2 cell (nm) - HP	808	864	832	576	840	696	576	456	384
Effective drive width of D1 ND2 cell (nm) - HD	101	126	128	96	120	96	84	72	72

4.4. SYSTEM-ON-CHIP (SOC) PPA METRICS

Due to standard cell and bitcell density improving on a node-to-node basis, it is possible to integrate more functions in a given SoC footprint. The footprint for mobile SoC integration is assumed to increase across generations because of newly added functions exceeding the scaling reduction. Therefore, the amount of memory as well as graphical processing unit (GPU) processors and neural processing units (NPU) follow the density scaling of SRAM and standard cell, respectively, and if the trend for more parallel architectures continues. Thanks to advances in DTCO, lateral nanosheets, followed by device-over-device stacking (e.g. P-over-N) and 3D VLSI, SoC footprint scaling factor for the same function can still be maintained. In this roadmap edition we also added more details on machine learning blocks such as the MAC (multiply-accumulate) block clock frequency and number of MACs in SoC.

Integration capacity of logic technology is shown in Figure MM-6 (amount of NAND2-equivalent standard cell density as well as bitcell density).

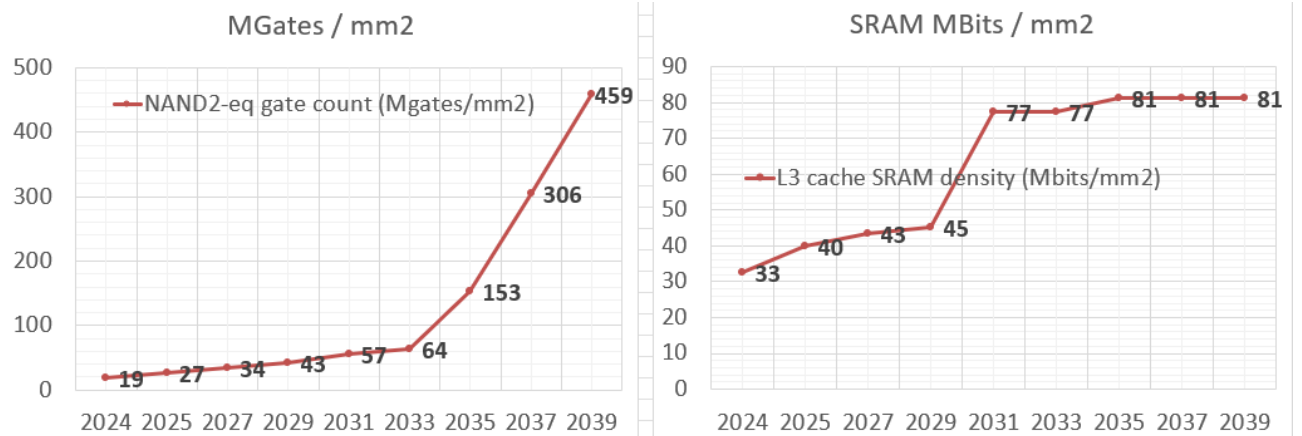
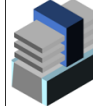
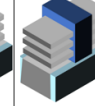
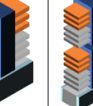


Figure MM-7 NAND2-equivalent standard cell count (left) and 1T1-bitcell (right) scaling.

Projected power and performance scaling of SoC is given in Table MM-11.

Table MM-11 Area, Power, and Performance Scaling of SoC

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fin-pitch 3D integration scheme	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	Stacking	Stacking	Stacking	CFET	CFET	3DVL SI	3DVL SI	3DVL SI
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
LOGIC TECHNOLOGY KEY CHARACTERISTICS									
Vdd (V)	0.70	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
Gate length (nm)	16	14	12	12	12	10	10	10	10
Number of stacked tiers [1]	1	1	1	1	1	1	2	4	6
Number of stacked nanosheets in PMOS/NMOS [1]	-	3	4	4	3	3	3	3	3
Cell height (nm) - HD	160	120	96	80	63	56	49	49	49
NAND2-eq gate count (Migates/mm ²)	19	27	34	43	57	64	153	306	459
L3 cache SRAM density (Mbits/mm ²)	33	40	43	45	77	77	81	81	81
Analog + IO scaling	1.00	0.90	0.81	0.73	0.66	0.59	0.53	0.48	0.43
SoC footprint scaling	1.00	0.77	0.67	0.59	0.42	0.39	0.28	0.24	0.23
#GPU cores in SoC - at integration capacity	36	51	64	80	106	119	285	571	856
#CPU cores in SoC - at throughput target, #CPUx/max	12	19	31	62	74	134	215	383	680
#MAC units in SoC - at capacity	16384	23211	29013	36330	48230	54259	129926	259851	389777
CPU frequency (GHz)	3.10	3.39	3.51	2.94	4.17	3.93	4.18	3.99	3.82
CPU frequency at constant power density (GHz)	3.10	3.14	4.14	3.02	4.07	3.60	2.00	1.13	0.79
MAC frequency (GHz)	1.80	1.97	2.04	1.71	2.42	2.28	2.42	2.32	2.22
MAC power scaling	1.00	0.76	0.60	0.55	0.50	0.47	0.38	0.32	0.29
MAC Power density scaling	1.00	1.08	0.85	0.97	1.03	1.09	2.08	3.55	4.81
MAC TOPS	1.0	1.6	2.0	2.1	4.0	4.2	10.7	20.4	29.3
MAC TOPS/mm ² scaling	1.0	2.2	3.5	4.7	11.7	13.9	84.7	323.8	696.6
MAC TOPS/W scaling	1.0	1.3	1.7	1.8	2.0	2.1	2.7	3.1	3.5
MAC TOPS/mm ² * TOPS/W	1.0	2.9	5.9	8.5	23.4	29.5	225.6	1014.0	2411.7

Clock frequency is projected to mildly improve because of increasing parasitics and limited gate drive ($V_{gs}-V_t$) as function of scaling. After 2035 3D-VLSI will help to reduce the parasitics through the wirelength reduction and through the split of cells in 3D. Also, thermal (increasing power density) constraints reduce the average frequency if the chip needs to be operated at constant power density. If nothing is done for the mitigation of thermal issues, the CPU needs to be throttled more frequently to maintain the same power density. The rate of power reduction tends to flatten because of slow-down in supply voltage (V_{dd}) and slow-down of capacitance scaling towards the end of roadmap. Impact of those trends in frequency, area, and energy to the system metrics such as area-efficient performance (TOPS/mm²) and energy-efficient performance (TOPS/W) is shown in Figure MM-8.

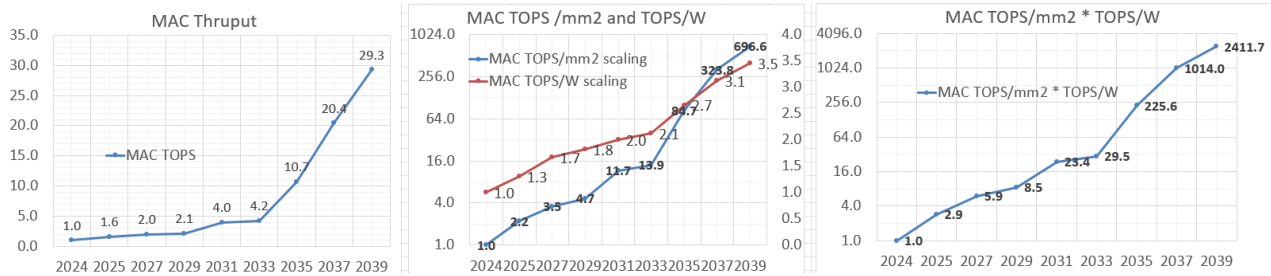


Figure MM-8 Scaling projection of MAC throughput, MAC TOPS/mm², and TOPS/W, and system area-and-power constrained performance metric (TOPS/mm²*TOPS/W).

4.5. INTERCONNECT TECHNOLOGY REQUIREMENTS

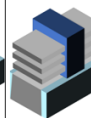
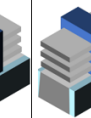
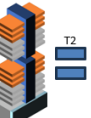
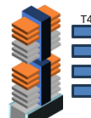
The most difficult challenge for interconnects is the introduction of new materials that meet the wire conductivity requirements, reduce dielectric permittivity, and meet reliability requirements. As for the conductivity, the impact of size effects on interconnect structures must be mitigated. Future effective κ requirements preclude the use of a trench etch stop for dual damascene structures. Dimensional control is a key challenge for present and future interconnect technology generations and the resulting difficult challenge for etch is to form precise trench and via structures in low- κ dielectric material to reduce variability in resistance-capacitance (RC). The damascene scheme used for integration requires tight control of pattern, etch, and planarization. To extract maximum performance, interconnect structures cannot tolerate variability in profiles without producing undesirable RC degradation. These dimensional control requirements place new demands on high-throughput imaging metrology for measurement of high aspect ratio structures. New metrology techniques are also needed for inline monitoring of adhesion and defects. Larger wafers and the need to limit test wafers will drive the

adoption of more in-situ process control techniques. Table MM-12 highlights and differentiates the top key challenges while Table MM-13 shows the interconnect scaling roadmap.

Table MM-12 Interconnect Difficult Challenges

Critical Challenges	Summary of Issues
Materials—Mitigate impact of size effects in interconnect structures	Line and via sidewall roughness, intersection of porous low- κ voids with sidewall, barrier roughness, and copper surface roughness will all adversely affect electron scattering in copper lines and cause increases in resistivity.
Metrology—Three-dimensional control of interconnect features (with its associated metrology) will be required	Line edge roughness, trench depth and profile, via shape, etch bias, thinning due to cleaning, CMP effects. The multiplicity of levels, combined with new materials, reduced feature size and pattern dependent processes, use of alternative memories, optical and RF interconnect, continues to challenge.
Process—Patterning, cleaning, and filling at nano-dimensions	As features shrink, etching, cleaning, and filling high aspect ratio structures will be challenging, especially for low- κ dual damascene metal structures and DRAM at nano-dimensions.
Complexity in Integration—Integration of new processes and structures, including interconnects for emerging devices	Combinations of materials and processes used to fabricate new structures create integration complexity. The increased number of levels exacerbates thermomechanical effects. Novel/active devices may be incorporated into the interconnect.
Practical Approach for 3D—Identify solutions that address 3D interconnect structures and other packaging issues	Three-dimensional chip stacking circumvents the deficiencies of traditional interconnect scaling by providing enhanced functional diversity. Engineering manufacturable solutions that meet cost targets for this technology is a key interconnect challenge.
Growing gap between area-efficient signaling and power distribution in the local interconnects	Increasing trade-off between area-efficient wiring within the cell versus low-resistance power delivery to the standard cell, necessitating decoupling of power rails from the signal routing such as introduction of buried power rails.

Table MM-13 Interconnect Roadmap for Scaling

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
Logic industry "Node Range" Labeling	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Fine-pitch 3D integration scheme	~3nm" Enhanced	"2nm"	"1.4nm"	"1.1nm"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	Stacking	Stacking	Stacking	CFET	CFET	3D VLSI	3D VLSI	3D VLSI
Backside structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Platform device for logic	finFET	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
									
FRONT-SIDE INTERCONNECT TECHNOLOGY									
Number of Mx layers	2	2	2	2	2	2	2	2	2
Number of P40 layers	2	2	2	2	2	2	2	2	2
Number of P80 and <P720 layers	8	9	10	11	12	12	12	12	12
Number of P720 layers	2	2	2	2	2	2	2	2	2
Number of wiring layers - M1+Mx+ >P40	15	16	17	18	19	19	19	19	19
Mx - tight-pitch interconnect resistance (Ohms/um)	300	475	475	475	475	475	475	475	475
Mx - tight-pitch interconnect capacitance (aF/um)	270	270	270	270	270	270	270	270	270
Vx - tight-pitch interconnect via resistance (Ohms/via)	50.0	53.0	64.0	64.0	64.0	64.0	64.0	64.0	64.0
MP80 - 80nm pitch interconnect resistance (Ohms/um)	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
MP80 - 80nm pitch interconnect capacitance (aF/um)	198	198	198	198	198	198	198	198	198
VP80 - 80nm pitch interconnect via resistance (Ohms/via)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Aspect ratio - M0, M1, Mx, MP80, MP720	1.5-2.5	1.5-2.5	1.5-4.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0
Power rail layer	M0	M0	Backside	Backside	Backside	Backside	Backside	Backside	Backside
Power rail material	Co, W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru
Metallization - M0	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru
Barrier - M0	0.5nm TaNCo	0.5nm TaNCo	Cu	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless
Metallization - M1, Mx	Cu	Cu	Cu	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo
Barrier metal - M1, Mx	1.5nm TaNCo	1.0nm TaNCo	0.5nm TaNCo	SICOH	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)	SICOH (2.20-2.55)
Di-electrics k value - M0, M1, Mx	SICOH (2.70-3.20)	SICOH (2.70-3.20)	SICOH (2.70-3.20)	SICOH (2.70-3.20)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)	Airgap (1.0)
Metallization - >MP40	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu
Di-electrics k value - >MP40	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)	SICOH (2.40-2.55)

4.5.1. Conductor

Copper (Cu) is expected to remain to be the preferred solution for the interconnect metal, at least until 2029 while non-Cu solutions (e.g. Co and Ru) are expected to be used for the local interconnect (M0). On the other hand, due to limits of electromigration, the local interconnect (middle-of-line (MOL)), M1, and Mx levels will embed non-Cu solutions such as Cobalt (Co), particularly for the via, due to its better integration window to fill the narrow trenches on top of the EM performance as well as its lower resistance compared to Cu at scaled dimensions. Although a resistivity increase due to electron scattering in Cu or higher bulk resistivity in non-Cu solutions (e.g., Co) are already apparent, a hierarchical wiring approach such as scaling of line length along with that of the width still can overcome the problem.

4.5.2. Barrier Metal

Cu wiring barrier materials must prevent Cu diffusion into the adjacent dielectric but also must form a suitable, high quality interface with Cu to limit vacancy diffusion and achieve acceptable electromigration lifetimes. Ta(N) is a well-known industry solution. Although the scaling of Ta(N) deposited by plasma vapor deposition (PVD) is limited, other nitrides such as Mn(N) that can be deposited by chemical vapor deposition (CVD) or atomic layer deposition (ALD) have recently attracted attention. As for the emerging materials, self-assembled monolayers (SAMs) are researched as the candidates for future generation.

4.5.3. Inter-metal Dielectrics (IMD)

Reduction of the IMD κ value is slowing down because of problems with manufacturability. The poor mechanical strength and adhesion properties of low- κ materials are obstructing their incorporation. Delamination and damage during CMP are major problems at early stages of development, but for mass production, the hardness and adhesion properties needed to sustain the stress imposed during assembly and packaging must also be achieved. Difficulties associated with the integration of highly porous ultra-low- κ ($\kappa \leq 2$) materials become clearer, and air-gap technologies is the alternative path to lower the inter-layer capacitance. As the emerging materials, metal organic framework (MOF) and carbon organic framework (COF) could be advocated.

4.5.4. Reliability—Electromigration

An effective scaling model has been established in the earlier editions of roadmap where it assumes that the void is located at the cathode end of the interconnect wire containing a single via with a drift velocity dominated by interfacial diffusion. The model predicts that lifetime scales with w^2h/j , where w is the linewidth (or the via diameter), h the interconnect thickness, and j the current density. Whereas the geometrical model predicts that the lifetime decreases by half for each new generation, it can also be affected by small process variations of the interconnect dimensions. J_{max} (maximum equivalent DC current density) and JEM (DC current density at the electromigration limit) are limited by the interconnect geometry scaling. J_{max} increases with scaling due to reduction in the interconnect cross-section and increase in the maximum operating frequency. The practical solutions to overcome the lifetime decrease in the narrow linewidths have been discussed actively over the past years. Recent studies show an increasingly important role of grain structure in contributing to the drift velocity and thus the EM reliability beyond the 45nm node. Process solutions with Cu alloys seed layer (e.g., Al or Mn) have shown to be an optimum approach to increase the lifetime. Other approaches are the insertion of a thin metal layer (e.g., CoWP or CVD Co) between the Cu trench and the dielectric SiCN barrier and the usage of the short length effect. The short length effect has effectively been used to extend the current carrying capability of conductor lines and has dominated the current density design rule for interconnects.

4.5.5. Reliability—Time Dependent Dielectric Breakdown

Basically, the dielectric reliability can be categorized according to the failure paths and mechanisms as shown in Figure MM-9. While a large number of factors and mechanisms have already been identified, the physical understanding is far from complete. For instance, it is necessary to correctly account for LER, voltage dependence, etc in modeling TDDDB lifetime that directly impacts the estimate of V_{max} (or minimum dielectric spacing).

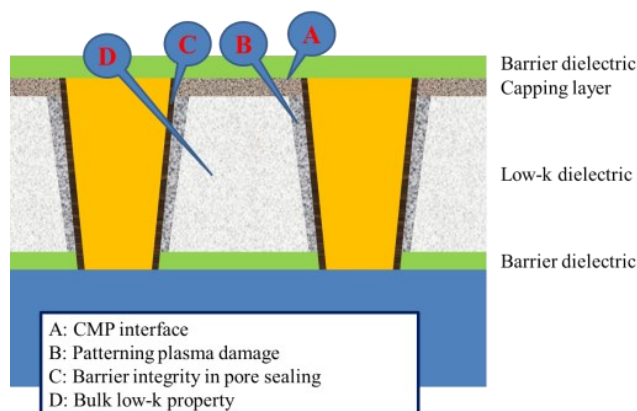


Figure MM-9 Degradation paths in low- κ damascene structure.

4.6. 3D INTEGRATION

Every logic generation needs to add new functions in each node to keep unit price constant (to preserve profit margins). This is getting more difficult due to the following challenges:

- Fewer functions left on board/system to co-integrate
- Heterogeneous cores specialized per function - specialized performance improvement requirements needed per each dedicated core
- Off-package memories - costly to co-integrate with logic, technology not compatible with baseline CMOS (where wafer/die-level stacking might be needed), limited bandwidth and energy efficiency to transport data from memory to logic

Die cost reduction has been enabled so far by concurrent scaling of gate pitch, metal pitch, and cell height scaling. This is expected to continue until 2031 and this slow-down has been accompanied by fine-pitch 3D stacking assembly such as uBump stacking and hybrid bonding [21]. Cell height scaling will continue to be pursued by DTCO constructs in cell and physical design. However, this scaling route is expected to be more challenged by diminishing electrical/system benefits and also by diminishing area-reduction/\$ at SoC level. Therefore, it is necessary to pursue 3D integration routes such as device-over-device stacking, fine-pitch layer transfer, and/or monolithic 3D (or sequential integration). These pursuits will maintain system performance and power gains while potentially maintaining the cost advantages such as treating expensive non-scaled components somewhere else and using the best technology fit per tier functionality. 3D stacking routes should factor in know-good-die sorting and test methodologies to improve the stacking yield where wafer-to-wafer stacking would require a very yielding process on each of the stacked wafers because of testing and wafer sorting challenges. Going more heterogeneity in die stacking would require careful planning how tiers are partitioned, for instance having a smaller I/O die on top logic die would require a lot of 2D routing in the logic die to fanin the connections from the corresponding logic block on the logic tier to the I/Os in the IO tier above. This routing would itself introduce the some area penalty in the logic tier. The overall trade-off should also include the assembly/stacking yield and additional wafer process steps such as TSV, wafer thinning, Cu pad/uBump processing.

3DVLSI can be routed either at gate or transistor levels. 3DVLSI offers the possibility to stack tiers enabling high-density contacts at the tier level (up to several million vias per mm²). The partitioning at the gate level allows IC performance gain due to wire length reduction while partitioning at the transistor level by stacking nFET over pFET (or the opposite), enabling the independent optimization of both types of transistors (customized implementation of channel material/substrate orientation/channel and raised source/drain strain, etc. [16]) while enabling reduced process complexity compared to a planar co-integration, for instance the stacking of III-V nFETs above SiGe pFETs [27]. These high mobility transistors are well suited for 3DVLSI because their process temperatures are intrinsically low. There is a significant momentum on integrating device-on-device stacking (e.g. N device over P) to decouple the channel engineering (e.g. Ge channel for PMOS) for better performance. Better performance of higher tiers achieved by a freedom to select a better substrate should however factor in the potential degradation of performance due to processing them at lower temperature budget compared to the devices at the most bottom tier.

In order to address the transition from 2D to 3DVLSI, the following generations are projected in the roadmap: